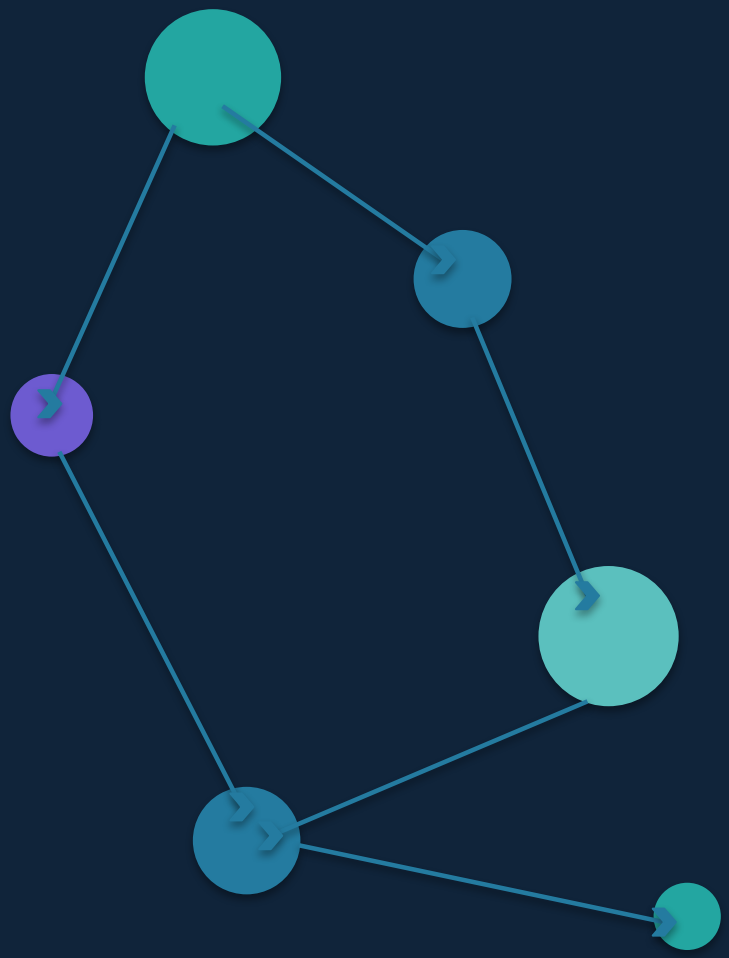
PROJECT ARCHITECTURE

GlucoTwin

System Architecture

A multimodal pipeline for glucose-risk prediction in Type 2 Diabetes

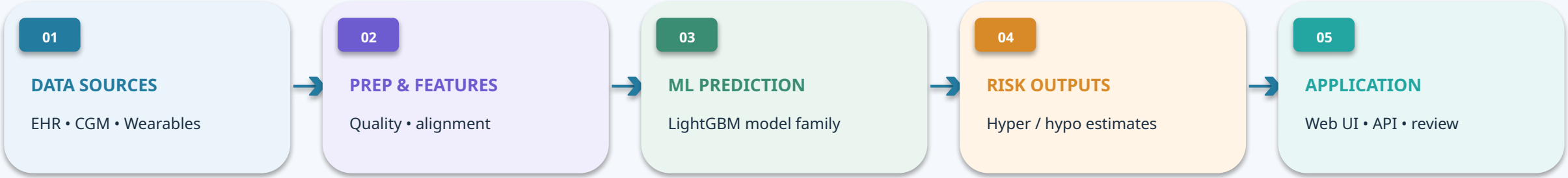
Patient data → feature engineering → ML inference → human-reviewed decision support



SUBMISSION EDITION • 20 OCTOBER 2026

A clear path from data to decision support

Clinical context and time-series signals feed risk estimates for human review.



INPUTS

Patient records
Glucose readings
Activity / sleep

PROCESSING

Data quality
Time alignment
Feature construction

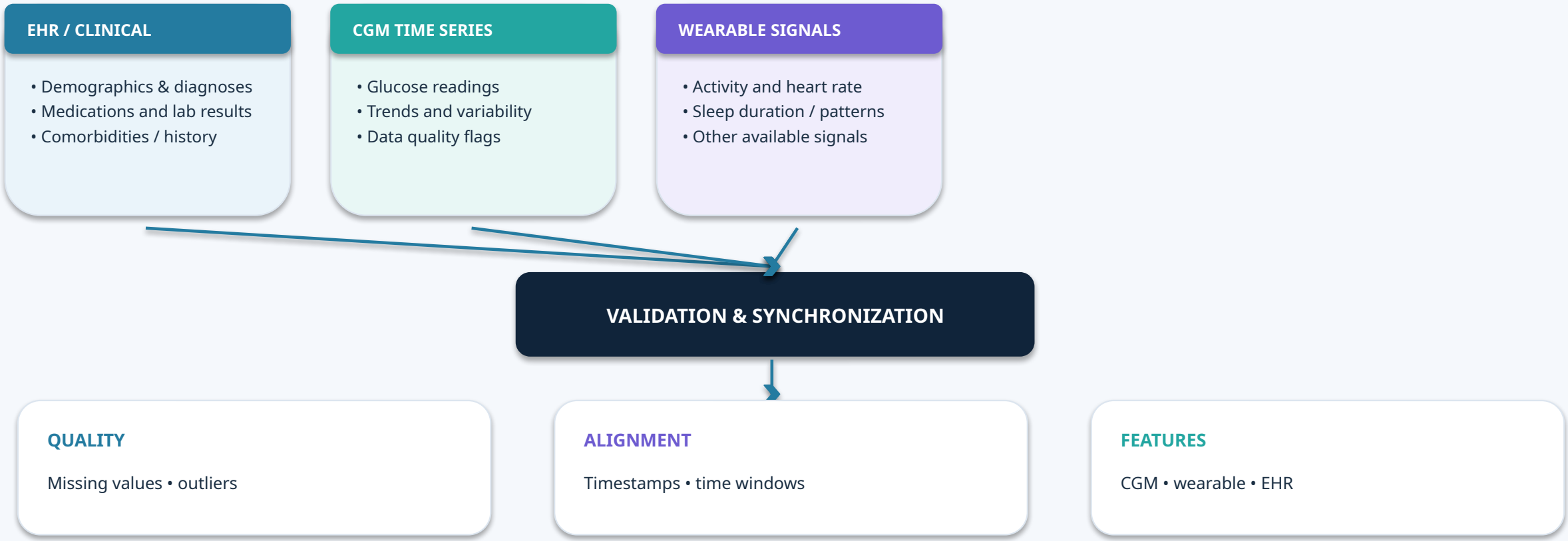
OUTPUTS

Risk estimates
Supporting context
Human review

Decision support only—not a diagnosis, prescription, or substitute for clinician judgement.

Three data streams become one model-ready view

The streams are validated and synchronized before feature construction.



Exact fields and time windows depend on the records available in the dataset.

Train, evaluate, and compare before use

LightGBM is the documented model family; evaluation design is as important as the algorithm.

MODEL INPUTS

Feature groups

- CGM-derived features
- Wearable-derived features
- EHR / clinical features
- Data-quality indicators

Use patient separation and preserve time order.

MODEL FAMILY

LightGBM

Separate prediction tasks:
- Hyperglycemia
- Hypoglycemia

Version model artifacts and record the evaluation setup.

EVALUATION

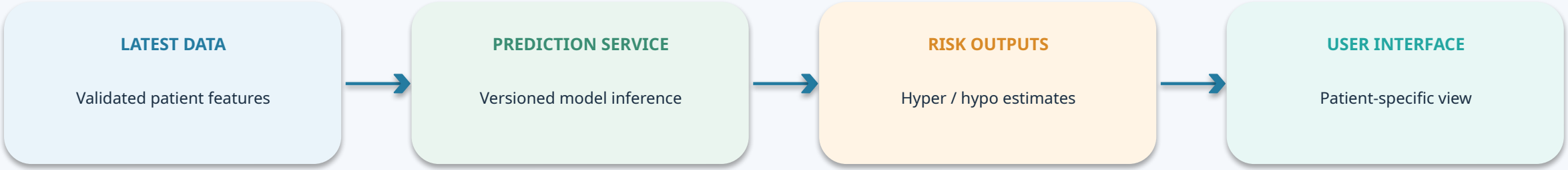
Measure multiple dimensions

- AUROC and AUPRC
- Brier score / calibration
- Sensitivity at a defined threshold
- Lead-time analysis
- Baselines and ablations

Report metrics only after verifying labels, split strategy, prediction horizon, and consistency across evaluation scripts.

Risk estimates are surfaced with context

The application presents model outputs for interpretation, not autonomous clinical decisions.



CLINICIAN REVIEW

Review estimates alongside history, current observations, data quality, and local clinical protocols.

EXPLANATION & LIMITS

Show supporting features only where available. Do not imply causal certainty or guaranteed outcomes.

Human-in-the-loop by design • No autonomous diagnosis or treatment recommendations

Operate with privacy, monitoring, and clear limits

Operational controls belong alongside the modelling pipeline.

APPLICATION LAYER

- React + Vite frontend
- FastAPI REST endpoints
- Patient-specific web views

DATA & MODEL LAYER

- PostgreSQL data store
- Python / LightGBM inference
- Optional Ollama / LLM guidance

SECURITY & PRIVACY

- Least-privilege access
- Protect identifiable health data
- Audit relevant access / changes

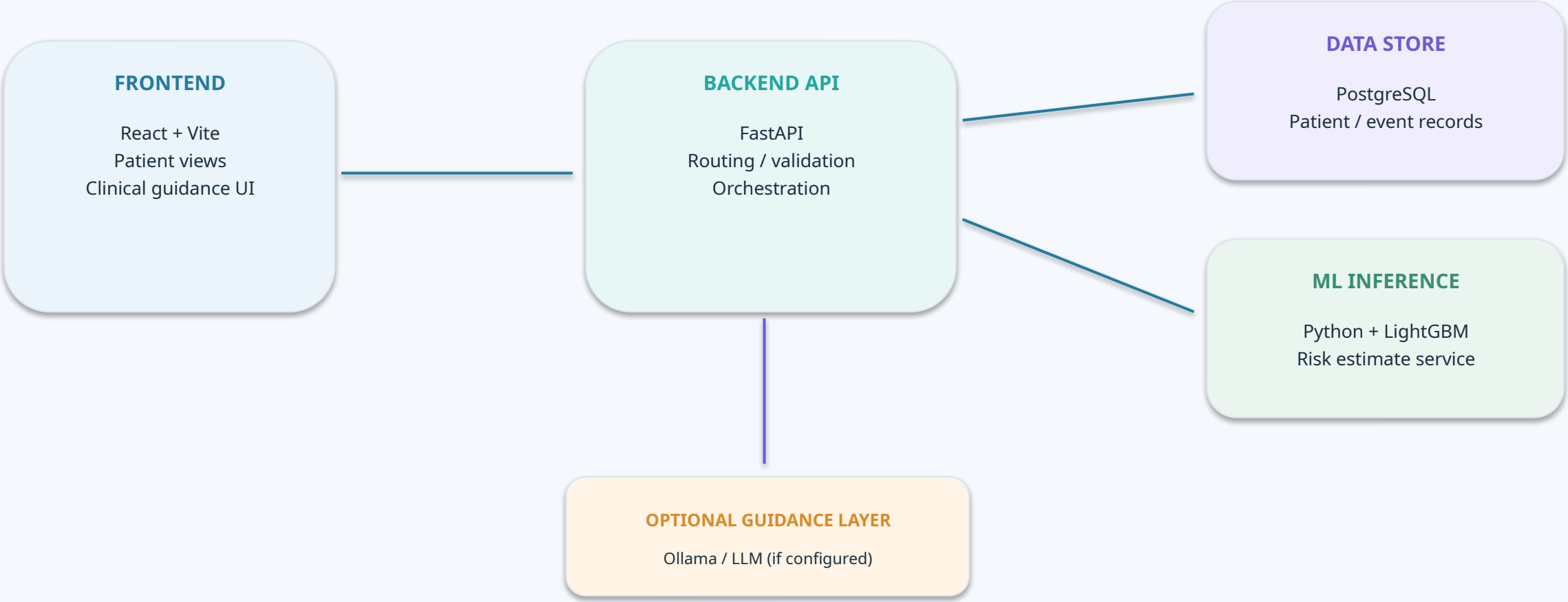
MONITORING & REVIEW

- Track data / model quality
- Re-evaluate before updates
- Clinical and safety review

Healthcare use requires validation, privacy safeguards, documented limitations, and qualified clinical oversight.

How the application components connect

Logical component view of the prototype stack; deployment boundaries may vary.



Keep credentials server-side; enforce authentication, authorization, input validation, and auditability before real-patient deployment.

Evaluation should be reproducible and transparent

Report model performance with baselines, clear test splits, and metrics appropriate for imbalanced outcomes.

DISCRIMINATION

AUROC • AUPRC

Can the model rank higher-risk cases above lower-risk cases?

CALIBRATION

Brier score

Do estimated probabilities align with observed outcomes?

OPERATING POINT

Sensitivity / specificity

What trade-off appears at the chosen threshold?

TIMING

Lead-time analysis

How early are events flagged under the defined horizon?

REPRODUCIBILITY CHECKLIST

Fixed seed • patient-level separation • chronological holdout • label / horizon definition • baseline comparison • versioned metrics

Do not combine metrics from different evaluation runs or present synthetic results as clinical validation.

Clear boundaries are part of the architecture

Healthcare AI must communicate uncertainty and preserve qualified human oversight.

DATA QUALITY

Missing, delayed, or inaccurate data can distort risk estimates.

GENERALIZATION

Performance can change across populations, devices, and care settings.

CLINICAL REVIEW

A qualified clinician must interpret outputs in patient context.

PRIVACY & SECURITY

Protect health data with access controls, secure configuration, and auditing.

INTENDED SCOPE

Prototype decision support only. Not a standalone diagnosis, treatment recommendation, or validated medical device.

GlucoTwin at a glance

A modular workflow connecting clinical records and time-series signals to glucose-risk estimates and human-reviewed context.

01	**Integrate**	EHR + CGM + wearable signals
02	**Prepare**	Validate, synchronize, engineer features
03	**Predict**	LightGBM model family
04	**Review**	Risk estimates with clinician oversight

CORE PRINCIPLE

Useful predictions require reliable data, transparent evaluation, and human judgement.

GlucoTwin • MedIntel AI / MedAI • Submission edition